

ATUL KUMAR

✉ Email | 🐙 Github | 🌐 Portfolio | 🔗 LinkedIn | 📄 Codechef

EDUCATION

Birla Institute of Technology, Mesra

Bachelor of Technology in Chemical Engineering

2023 – Present

Ranchi, India

Chinmaya Vidyalaya, Bokaro Steel City

Higher Secondary: Class: 11,12

2021 – 2022

Bokaro, Jharkhand

EXPERIENCE

Web Development Intern

Apr 2026 – Present

Zaalima Development (Remote)

- Developing a dashboard UI to manage and track 100+ AI-generated extensions, improving workflow visibility and user experience.
- Implementing an “Edit Request” feature enabling real-time modification of generated code via prompts.
- Building subscription-based feature gating for API-driven functionalities and actively deploying the subscription workflow.
- Integrating code sanitization pipelines and conducting ongoing security audits to prevent malicious outputs in AI-generated code.

PROJECTS

Automated Email Outreach System | *Python, Pandas, SMTP* 🐙

Feb 2026

- Developed a Python CLI tool to automate email outreach to 100+ HR using structured CSV data.
- Integrated secure Gmail SMTP with environment variable authentication to prevent credential exposure.
- Built data validation and preprocessing pipeline to handle missing/invalid entries.
- Implemented rate-limited batch delivery with error handling, reducing manual effort by 90%.

Dental Caries Segmentation | *YOLOv8, PyTorch, Computer Vision* 🐙

Jan 2026

- Developed a YOLOv8-based instance segmentation pipeline for detecting dental caries from **750+** radiographic images.
- Trained on 768×768 images using AdamW optimizer with cosine learning-rate scheduling.
- Applied augmentation techniques to increase effective dataset diversity by approximately 4×.
- Achieved 22.6% mask recall using Precision, Recall, and mAP evaluation metrics.

Market Regime Detection | *Python, Pandas, NumPy, Scikit-learn* 🐙

Dec 2025

- Analyzed 7 years of NIFTY 50 data across **1,700+** trading sessions.
- Engineered 10+ market indicators, including returns, volatility, and momentum features.
- Applied clustering algorithms to identify **3** market regimes: Bull, Bear, and Sideways.
- Discovered regime-specific return-volatility patterns for improved market interpretation.

TECHNICAL SKILLS

Programming: Python, C, C++

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Model Evaluation

Libraries: Scikit-learn, NumPy, Pandas, PyTorch

Databases & Tools: MySQL, DBMS, Power BI, MS Excel, Tableau

Technologies: Git, Docker, Jupyter Notebook

Web Development: HTML, CSS, JavaScript, WordPress

Cloud: AWS, Google Cloud

CERTIFICATIONS & JOB SIMULATIONS

Deloitte Data Analytics Job Simulation (Forage)

Data Analyst Bootcamp (Udemy)